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FREIE UNIVERSITÄT BERLIN

Measuring Air Quality in Urban Environments

An applied mobile citizen science methodology

Introduction

The advancement of sensor technology, the availability of affordable small computing units and the increasing capabilities of Large-Language-Model Chatbots made the gathering of environmental data broadly accessible [1,2] .While not having the highest sensor accuracy, spatial and time related data resolution of, among other, air quality, can be dramatically increased through providing fine-scaled public data to public platforms by citizen scientists [3]. This project attempts to apply air quality with a self-assembled sensor kit, create software for formatting and interpolating the data, ultimately creating map and graph material to visualize it.

3D-Printed Case
Protective custom enclosure used starting 1 Feb 2026 [4].

A Raspberry Pi Zero 2 W was used as computing unit for the sensors.

Pimoroni Enviro+ board, Measuring temperature, humidity, pressure, presence of oxidizing gases like ozone (O_3) and nitrogen dioxide (NO_2), reducing gases such as nitric oxides (NO) and specifically ammonium NH_3[5,6].

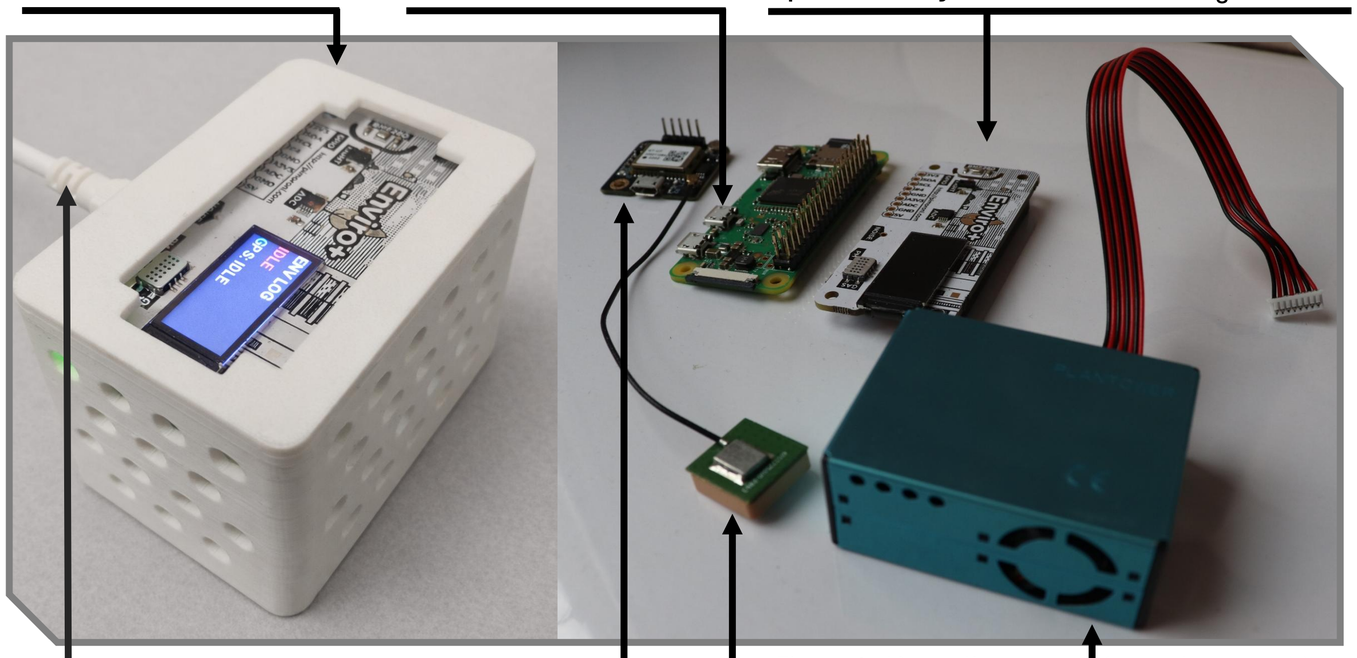

Figure 1. Air quality sensor (own photograph).

Can be powered by any **Powerbank** and Powersource of 5 Volts.

GT-U7 GPS Receiver & Ceramic Patch Antenna
Provides Coordinates and timestamp Connected via USB to avoid UART conflicts (micro-USB ↔ USB-A adapters).

PMS5003 Sensor
Measures particulate matter in PM1.0 / PM2.5 / PM10. Connected via UART through the Enviro Plus board.

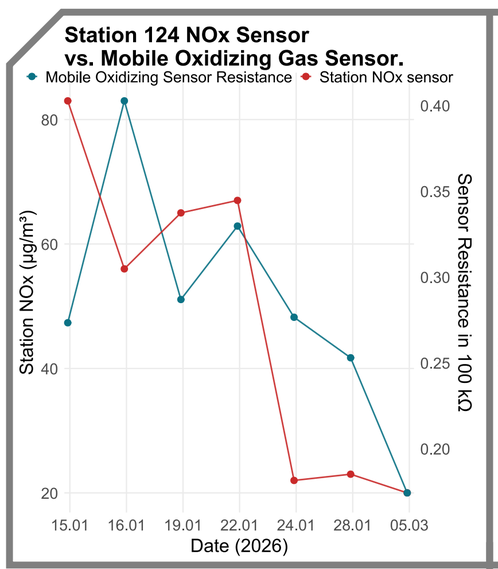

Figure 2 – Oxidising gasses vs. NOx

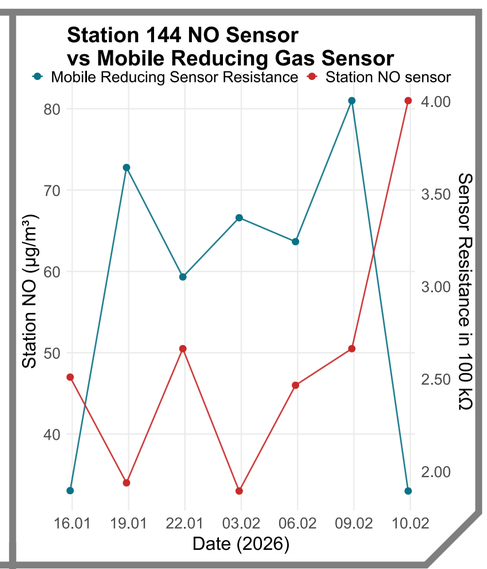

Figure 3 - Reducing gasses vs. NO

Figure 2 and 3 show two alternating air quality measurement station [7] passes (location in Fig. 4) with values in blue showing mobile measurements of oxidzing and reducing gases (see top right of poster).

Figure 4 shows a map containing averaged values for oxidizing and reducing gasses, and particulate matter (2.5) along bike-routes.

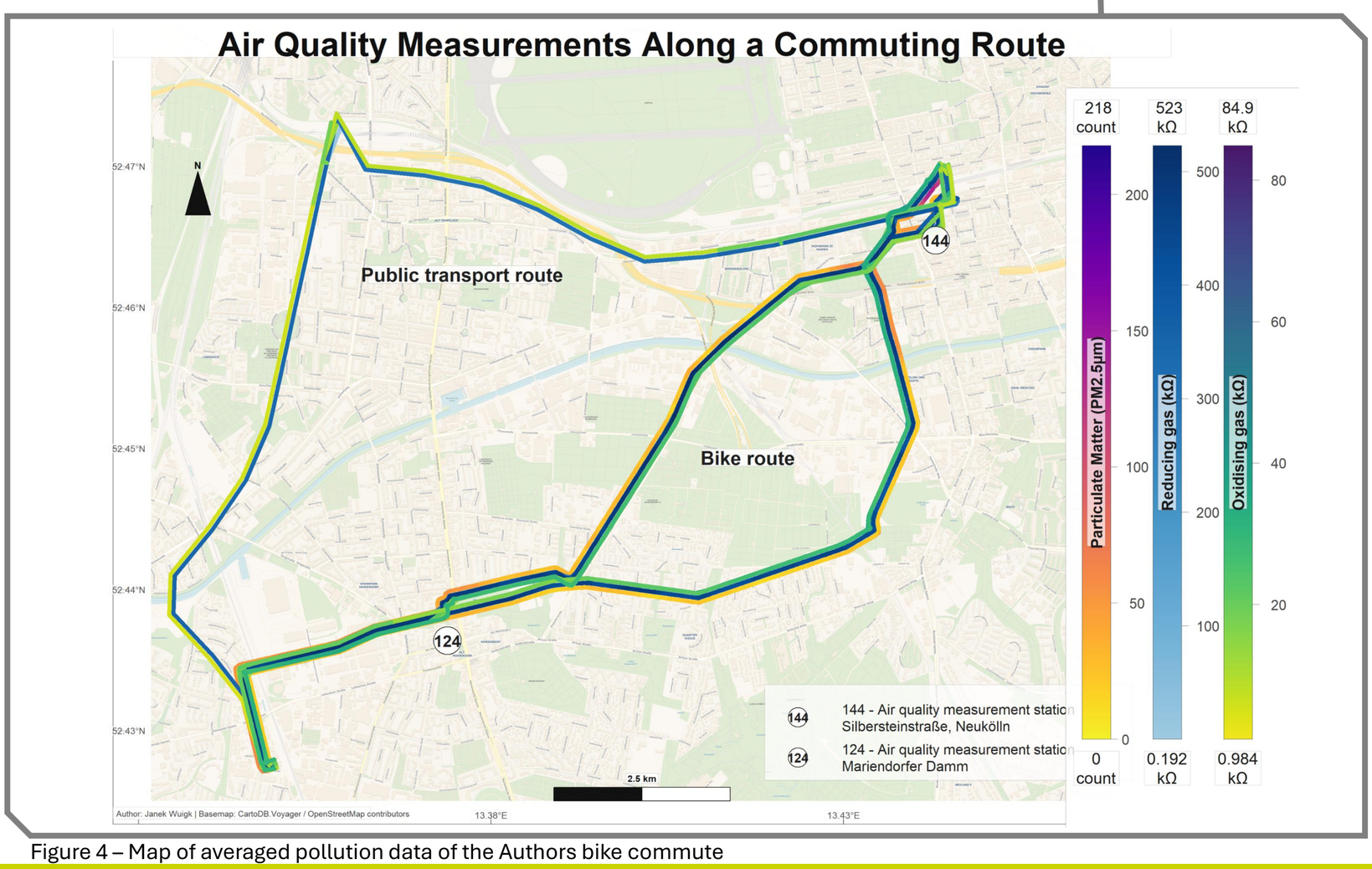

Figure 4 – Map of averaged pollution data of the Authors bike commute

Results and Outlook

The experience of recording the data in the field showed that while the sensors from the Enviro+ Chip recorded consistently and reliably, there were issues with catching gps signals and continuous particulate matter recordings. This required extensive interpolation of resulting datasets. As Fig. 2 shows, the data of the oxidizing gas sensors was not indicative of patterns of publicly recorded gases of those groups, just allowing very rough scale estimations. Figure 3 shows an inverse relationship, for reducing gases. The created software workflows for data interpolation and visualization might prove useful for deployment in similar, more accurate sensors with more accurate sensors and common gps issues [8,9].
It could further help potential citizen scientists, like the reader of this poster, to reproduce and improve the code posted under the Github documentation linked by the QR code.

Literature:

AI use statement: Generative artificial intelligence (AI) tools were used to support aspects of this project, including research assistance, programming, and language editing (spelling and orthography). Specifically, models from the GPT-5.2 to GPT-5.4 series were used between January 2026 and the end of March 2026.

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[2] Guo, ., et al. (2025). *AgentSense: LLMs empower participatory urban sensing.* ariv.
[3] Bagkis, E., Hassani, A., Schneider, P., DeSouza, P., Shetty, S., Kassandros, T., ... & Khan, J. (2025). Evolving trends in application of low-cost air quality sensor networks: challenges and future directions. *npj Climate and Atmospheric Science*, *8*(1), 335.
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[9] Mears M, Brindley P, Barrows P, Richardson M, Maheswaran R. Mapping urban greenspace use from mobile phone GPS data. PLoS One. 2021 Jul 7;16(7):e0248622. doi: 10.1371/journal.pone.0248622. PMID: 34232961; PMCID: PMC8262795.

Github Repository

github.com/GeoCodek/EnviroCommute